

From Data to Solutions

An AI workshop, Summer 2026

Carlos Cotrini and the teaching team

MAS AID / CAS DML / CAS AIS ETH Zürich



SECTION 1

About us

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About Carlos



Dr. Carlos Cotrini

- BSc Mathematics (2009)
- MSc Information Security (2014)
- PhD Information Security (2019)
- PostDoc in AI and ML (2019 to 2021)
- Senior Scientist, ETH (2021 to now)

Teaching at ETH and UZH

- Advanced Machine Learning
- Statistical Learning Theory
- AI in Medicine
- Data Analysis in Physics
- Stochastics and Machine Learning
- Building AI/ML Applications
- AI/IT in Industry
- Software Engineering Fundamentals

We study how to teach this, too

- This workshop is built on education research.
- **Full-stack ML**: build models from the ground up, not library-first.
- In a controlled trial, it beat the library-first approach by about 10% on conceptual understanding ($p = 0.006$), with more curiosity.

Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction

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Abstract

Balancing conceptual depth with practical skill development is a persistent challenge in advanced machine learning (ML) education, where powerful frameworks can obscure underlying mathematical and computational principles. To address this, we defined a new principled approach that we call full-stack machine learning (FSML), which emphasizes the construction of large language models and diffusion models from scratch. To evaluate the effectiveness of FSML, we conducted a classroom-based randomized controlled trial (N=288) in which FSML-based instruction was compared against a popular library-based instructional approach. We measured students' conceptual understanding through a specialized assessment and administered a survey capturing knowledge-gap awareness, curiosity, and cognitive load. We found that students who received FSML instruction performed approximately 18% better than control participants in a quiz on transformers and stable diffusion ($p = 0.006$). They also showed increased curiosity and more positive affective responses, suggesting deeper engagement with ML fundamentals. Our findings indicate that our full-stack approach to ML education can improve student learning outcomes, potentially reducing cognitive load for ML and other advanced computing topics.

ACM Reference Format

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1 Introduction

Computer science (CS) education face many dilemmas that are familiar to their colleagues from other disciplines, one of which is how to balance high-level conceptual approaches to the subject matter with foundational skill development. While top-down approaches that involve authentic and meaningful tasks can improve the motivation and situational interest of students [3, 12], they may leave students short of practical proficiency when not paired with consistent practice in drill-work—a problem well known, for example, in fractional literacy education [15, 21] and mathematics [13, 31]. Consequently, many educators favor a blend of top-down and bottom-up methods to capitalize on the strengths of both, although the best way to integrate them in computing education remains a topic of ongoing discussion [39].

In machine learning (ML) education, this question is particularly

Transforming Confusion into Diffusion (2025)

Pathos, to appear at ICER

In preparation: dynamic programming, diffusion models, RL.

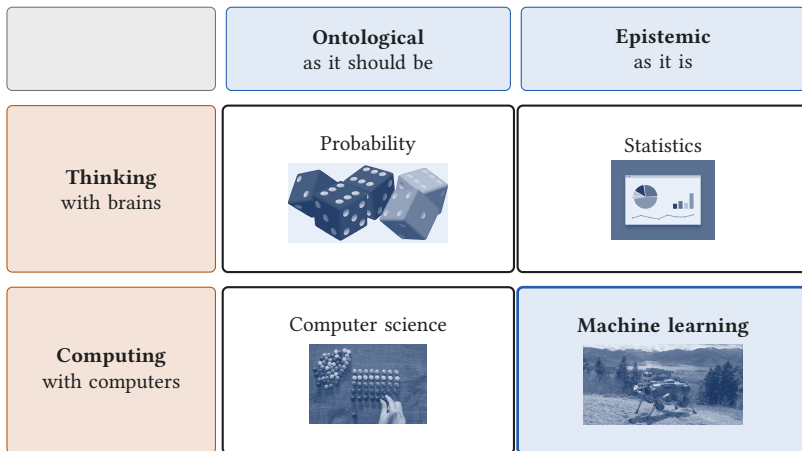


SECTION 2

What is AI?

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Four paradigms in science



Artificial intelligence: a working definition



Systems that perform tasks normally requiring **human intelligence**.

Main approaches:

- **Symbolic:** rules and logic
- **Machine learning:** estimate from data
- **Deep learning:** deep neural networks

Artificial intelligence: a timeline

Foundations



Turing test (1950)



Dartmouth (1956)



Perceptron (1958)

Expert systems, AI winter

LISP (1958)

DENDRAL, MYCIN (1970s)

Statistical ML

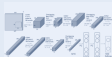


Deep Blue (1997)



IBM Watson (2011)

Deep learning



AlexNet (2012)

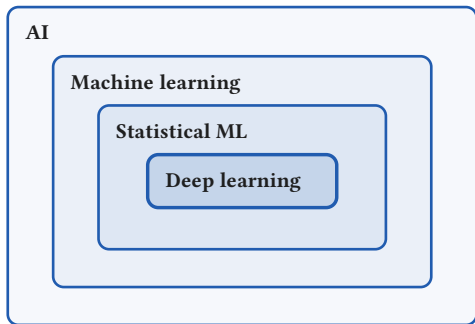


AlphaGo (2016)



ChatGPT (2022)

From AI to deep learning



- **AI:** act intelligently, by whichever means
- **ML:** learn the rules from data
- **Statistical ML:** minimize risk on data (Vapnik)
- **Deep learning:** with deep neural networks

From data to solutions.

You will learn to turn data into AI systems that create real value.

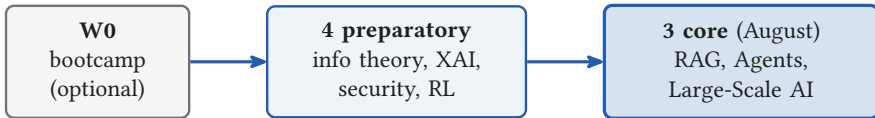


SECTION 3

The journey: eight weekends

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The shape of the workshop



everything builds toward the August core

The eight weekends

Bootcamp		4 preparatory				3 core, the August block		
W0 Python	W1 Info theory	W2 XAI	W3 Security	W4 RL	W5 RAG	W6 Agents	W7 Scale	
Jun 26	Jul 10	Jul 17	Jul 24	Aug 7	Aug 14	Aug 21	Aug 28	

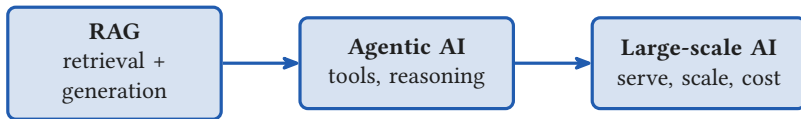
Each weekend: Friday (full day) and Saturday (morning).

Preparatory weekends: your toolbox

- **Information theory and competitive ML:** Gaussian processes, Bayesian optimization, Kaggle-style competition.
- **Explainable AI and LLM optimization:** SHAP, mechanistic interpretability, prompt optimization.
- **Security and AI:** adversarial attacks, jailbreaks, differential privacy.
- **Modern reinforcement learning:** REINFORCE, GAE, PPO.

four weekends of skills the August core will use

Core weekends: what you ship



- The three August weekends are the **mandatory core**.
- Each builds on the one before: a system you grow, then scale.

The projects



- Every weekend has one. You **submit four**: two preparatory, two core.
- Done **in pairs**, assessed **pass or fail**.
- Industrial framing, and a number on every demo (accuracy, win-rate, cost, latency).
- Choose your TA on Moodle, and defend each project before the grand deadline.



SECTION 4

Administration

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Passing the workshop

Please read these slides carefully.

- **Quizzes:** a short quiz after each preparatory and core weekend; retake until you reach at least 80%. The bootcamp has none.
- **Projects:** submit four (two preparatory, two core), in pairs, pass or fail.
- **Defense:** defend each project to a TA in a 15-minute Zoom presentation; **register** each defense on the course website.

All four projects are due
Saturday, 31 October 2026, 23:59
one grand deadline for the whole module

Attendance and recordings

- In-person attendance is **optional** for every weekend.
- All lectures are **recorded**: attend live or catch up later.
- The bootcamp (W0) is entirely optional.



recorded

Office hours and the course website

- The teaching team is available **virtually between weekends**.
- Drop by to get unstuck on a lecture or a project.



office hours

Office-hours schedule, Zoom link, and sign-up are on the course website.

Course website · office hours & project-defense registration

<https://eth-fdd-fs26.github.io/FDD-Page/#/>

Questions?